

CHETANA MURALIDHARAN

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EDUCATION

San Jose State University

M.S., Applied Data Intelligence GPA: 3.88/4.00

San Jose, CA

August 2024 – May 2026

Visvesvaraya Technological University

B.E., Computer Science & Engineering CGPA: 8.5/10.00

KA, India

August 2017 – August 2021

TECHNICAL SKILLS

Languages: Python (primary), SQL, Java

LLMs & Agentic AI: LLMs, Agentic Workflows, Multi-Agent Systems, On-Premises LLM Deployment (Ollama), Structured Output Validation, Prompt Engineering, RAG, GraphRAG, MCP, Semantic Search, Fine-Tuning (LoRA, PEFT), LangChain, LangGraph

ML and Deep Learning: PyTorch, TensorFlow, Transformers, Conformer-CTC, Multimodal AI, Model Benchmarking and Evaluation, Approximate Nearest Neighbor Search, Diffusion Models, Contrastive Learning, Knowledge Distillation, GANs

Inference and GPU: GPU-Based Training and Inference, Mixed Precision Training, Inference Optimization (Quantization, kv-cache), Data Parallelism, Nvidia Container Toolkit, GPU Profiling (nvprof, nsys)

Data Engineering: ETL/ELT Pipelines, CDC Pipelines, HDF5, Schema Validation, Apache Spark, Kafka, Talend, Snowflake

Databases & Storage: AWS S3, MongoDB, MySQL, Oracle, Graph Databases, Vector Databases (FAISS, Pinecone)

Software Engineering: FastAPI, REST APIs, RectFlow, Docker, Docker Compose, Git, Linux, CI/CD

EXPERIENCE

Instructional Student Assistant(ISA) - DATA 266: Generative Models

January 2026 – May 2026

San Jose State University

San Jose, CA

- Designed and maintained 14 assignments and 2 labs spanning Transformer architecture, diffusion models, LoRA, RAG, multimodal agents, CUDA programming, and inference optimization, building a curriculum that bridges theory to deployable AI systems.
- Debugged 100+ student AI workflows and multimodal PyTorch pipelines, while guiding LLM application development using Hugging Face models, vector databases, and agentic frameworks, improving end-to-end project reproducibility.

Full Stack Engineer Analyst

August 2021 – July 2024

Accenture

India

- Reduced manual data preparation effort by designing and deploying 60+ ETL/ELT pipelines using SQL, Talend, Snowflake, and AWS S3, delivering analytics-ready datasets for enterprise reporting.
- Cut data processing latency from 6 hours to 45 minutes by re-architecting SQL transformations and implementing CDC-based incremental ingestion pipelines, enabling near-real-time KPI reporting for operational decision-making.
- Increased data reliability from 85% to 96-98% by implementing schema validation, reconciliation checks, and automated monitoring frameworks, reducing downstream reporting errors across cross-functional teams.

PROJECTS

WeaveNet - LLM-Powered Graph Retrieval & Semantic Search Engine

August 2025 – May 2026

- Built an enterprise email intelligence platform over 517K emails enabling analysts to explore communication patterns, topic relationships through graph traversal, transforming unstructured emails into a knowledge graph of 764K nodes and 2.86M edges.
- Ran Qwen2.5-14B via HuggingFace with structured JSON output validation, collapsing 121K LLM-extracted topics into 5K canonical graph nodes(BAAI/BGE embeddings, HDBSCAN clustering), eliminating topic fragmentation across retrieval.
- Benchmarked 8 ANN algorithms (HNSW, ScaNN, LSH, Annoy, & variants) across 40 query types achieving 0.943 Recall@K and sub-100ms latency with FastAPI backend, custom query language, and React Flow frontend.

GuardNet - Multi-Modal Video Analytics & LLM Incident Summarization

August 2025 – December 2025

- Outperformed a CNN-LSTM baseline by 31% accuracy by building a MobileNetV2-BiLSTM pipeline with Multiple Instance Learning, achieving 92.19% accuracy and 0.9648 ROC-AUC on curated lab footage, real-world CCTV footage across stores.
- Reduced redundant GPU computation by 60% while maintaining 96.9% recall by replacing dense uniform frame sampling with motion-energy-guided extraction of the top-3 peak-motion clips per video.
- Integrated Mistral LLM to convert structured YOLOv8 object detection and pose-estimated wrist-to-pocket event logs into timestamped natural-language incident summaries, replacing manual video review with an explainable automated pipeline.

DocuLens - Multimodal RAG for Document Intelligence

August 2025 – December 2025

- Built a document question-answering system that retrieves text, figures, and tables from PDFs using hybrid BM25, FAISS retrieval (BAAI/bge-large-en-v1.5), RRF fusion, HyDE, reranking, enabling analysts to query reports without manual document review.
- Enforced hard citation rules in the generation layer, producing structured CSV outputs with zero hallucinated figure references across all 15 evaluation questions, making the system reliable enough for factual document intelligence use cases.

AgentLens - Multimodal Multi-Agent Image Understanding & Generation

August 2025 – December 2025

- Built a 4-agent image transformation system (visual understanding, prompt enrichment, image generation, and CLIP-based quality critique) that eliminated manual prompt engineering entirely and produced structured accept/revise decisions across all test cases.
- Implemented graceful fallback logic at each agent boundary, reverting to original user prompt on refinement failure and logging errors without pipeline crash, ensuring stability across ambiguous inputs and poor-quality images.